River Deep Mountain AI Use Case

Use Case 4: How might we identify water quality risk spots in a catchment with intensive agriculture?

River Deep Mountain AI

Version overview

Version	Date	Authors	Comments
0.1 Draft	January 2026	Richard Gooday (ADAS), Lionel Delaney (ADAS)	Comments from WRc, NWG, ADAS, Cognizant and the Rivers Trust addressed.
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1.0 Final Draft	February 2026	Richard Gooday (ADAS), Lionel Delaney (ADAS)	Final version of report.
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Approval

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Executive Summary

This report outlines a use case demonstrating how parts of the River Deep Mountain AI (RDMAI) model suite, specifically the Open Risk Map, Open Bare Cropland Detection, Open Poached Land Detection and Open Slurry Tank Detection models could be applied to support the identification of water quality risk spots in catchments with intensive agriculture.

Agriculture is often a significant contributor to water quality issues, with legislative requirements to reduce this contribution (Environment Act, 2021) and water companies interested in reducing pollution at source due to cost-efficiencies compared with more expensive treatment options. However, given the large proportion of land used for agriculture in many catchments, and the diffuse nature of the associated pollution, it is often hard to identify where within a catchment it would be best to try and mitigate the potential causes of agricultural pollution. The RDMAI models bring together differing datasets to highlight potential areas of risk and satellite imagery to help identify features typically associated with greater risk of agricultural pollution, which together could be used to help target sites for further on-the-ground investigation and ultimately pollutant mitigation.

The report outlines the relevant RDMAI models and required datasets, and considers the value proposition of applications, as well as barriers to adoption and model limitations. The workflow is demonstrated by applying the models to the Smallhope Burn waterbody, which is a tributary of the River Browney in County Durham, a focus area for the Thriving Catchments project, led by Northumbrian Water and The Rivers Trust.

1 Identifying water quality risk spots in a catchment

It is estimated that 50% of nitrate pollution, 25% of phosphorus pollution and 75% of sediment pollution in the water environment comes from agriculture (Defra, 2018). Pressure from agriculture affects 40% of our water bodies, as determined by Reasons for Not Achieving Good Status (RNAGS) as defined by the Wate Framework Directive (Environment Agency, 2021). The Environment Act of 2021 has set a legally binding target to reduce nitrogen, phosphorus, and sediment pollution from agriculture into the water environment by at least 40% by 2038.

The Government has developed both regulatory and incentive approaches to try and reduce agricultural pollution. For example, Catchment Sensitive Farming (CSF) is a joint program between Natural England, the Environment Agency, and Defra to engage with the agricultural industry to voluntarily reduce agricultural water pollution in England. It was launched in 2006, targeting specific high-risk 'priority' catchments, but has since evolved into a national program. Defra's agri-environment schemes (most recently the Environmental Land Management Schemes (ELMS)) can provide funding for actions which are intended to improve water quality. The Farming Rules for Water (FRfW) 2018 are the most recent regulations in England designed to reduce agricultural water pollution, requiring farmers to plan nutrient applications, protect watercourses, and prevent soil erosion.

However, water pollution is also an issue for water companies. They often use 'Catchment Management' to prevent pollution from getting into raw water sources, rather than relying on 'end of pipe' solutions (like water treatment facilities) which are generally significantly more expensive. Other stakeholders, such as river trusts and wildlife organisations, are also often engaged in catchment management, either due to the direct benefits to water quality, or the co-benefits such as habitat creation.

One of the key factors is the identification of the best areas to visit or farms to engage with in order to encourage implementation of mitigation options and changes in farm practices, where they can achieve the greatest and/or most cost-effective reductions in pollution. This typically involves the identification of intensive agriculture, or areas of high risk, including those with good connectivity to watercourses.

1.1 Existing approaches

The CSF scheme is designed to be targeted at higher risk farms, and uses a bespoke modelling system based on farm data available within government, to rank farms according to land use and livestock and their associated pollutant risks. However, this risk ranking does not take account of whether the pollutant risk actually translates to a water quality risk in the river. CSF officers are able to diverge from this ranking

where additional information (such as that below) indicates alternative farms may be higher risk.

For other stakeholders who wish to identify water quality risk spots / high risk agricultural activities, there are a range of approaches, tools and datasets that can be used to determine locations of interest. Some of these are:

1. Catchment walk overs - a field-based assessment where specialists walk along watercourses and/or travel around the surrounding land to identify pollution sources, environmental pressures, and habitat conditions. It can also include the use of drones, which can increase the speed of surveying, provide additional information or allow surveying of less accessible areas.
2. Agricultural Land Environmental Risk and Opportunity Tool (ALERT), which contains data layers such as land use, topography and satellite imagery, and was designed to help locate interventions to reduce pollution from agriculture.
3. MAGIC in England or DataMapWales, which contain geographic information about the natural environment, land designations, scheme eligibility etc. from across government, to understand the water quality status and pollution issues within an area.
4. Catchment Data Explorer contains the WFD designations for each catchment in England, alongside any Reasons for Not Achieving Good Status (RNAGS).
5. Catchment Based Approach Data Hub a catalogue of data, maps and applications suitable for supporting catchment management planning
6. LandIS to identify soils, as pollutant issues can vary by soil type (e.g. lighter texture soils will have a higher nitrate leaching risk than heavier soils, and heavier soils are more prone to waterlogging).
7. Google Earth imagery can be used to inspect the catchment and identify areas that may be higher risk for pollution and should be targeted for inspection.
8. SCIMAP is a spatial framework for considering where diffuse (agricultural) pressures originate in the landscape and, therefore, where mitigation actions would be most effective.
9. High-resolution water quality monitoring – typically water company or citizen science data at sub-catchment scale that can show where potential pollutant sources are located.
10. Land use data, such as the Crop Map of England (CROME) data or UKCEH land use products, which can identify higher risk crops.

11. SAGIS is a GIS based source apportionment system that provides a spatial breakdown of chemical inputs to rivers and lakes from different contributing sectors.
12. E-planner a tool developed by UKCEH to help farmers and other land mangers identify the most suitable places for different environmental management actions via interactive maps.
13. NatureInsight a tool to identify suitable nature-based solutions that can address flooding, sequester CO_2 and restore habitats.

Catchment Data Explorer and MAGIC are more useful for identifying which catchments (down to WFD waterbody scale) are of interest, but cannot really be used for more detailed investigation. Catchment walk overs are generally the best method to identify high risk areas, but are labour intensive and can miss transient features and activities not there when the walk over is being undertaken (such as areas of bare soil or manure spreading). Some of the other items in the list are helpful for targeting where to focus any walk over so that time is used most efficiently. High resolution monitoring is really the only way to determine water quality risk, with the other items then useful for helping estimate the potential causes of any issue. The SAGIS tool allows the user to determine the sectoral contribution to pollutant loads along a watercourse, although the agricultural sector is assumed to be the 'gap' between the contributions from known sectors and the observed concentrations, rather than being explicitly modelled or predicted. E-Planner and NatureInsight are examples of tools developed to help more optimally locate mitigation measures and nature-based solutions, although they do not have an explicit pollution focus (but do account for some features relevant to sources and pathways for water pollution).

There are thus already a variety of tools and data that can be used to help identify water quality risk spots and where agriculture may be of particular concern. However, some of these are not easily available (e.g. SAGIS) or need a licence (UKCEH land use data) and there would often be limited consistency in the approach and results of using them between different people and organisations.

2 Using AI and machine learning to identify water quality risk spots

This use case demonstrates how four of the eight RDMAI models can be used to help identify areas of intensive agriculture and agricultural management that can potentially lead to increased pollution risk, and thus where changes in management could have a greater effect on water quality. The RDMAI Open Risk Map identifies where in the catchment agriculture might be the most important factor for water quality, whilst the Slurry Tank detection model finds one of the features associated with intensive agriculture (and thus potentially greater pollutant risk) and the Bare

Cropland and Poached Land detection models identify areas that can be associated with higher risk of pollutant mobilization.

These four models are briefly described below, further details on each of them can be found in their respective model documentation.

2.1 Open Risk Map

The Open Risk Map is built up of individual risk layers, each of which aligns with a particular pollutant (e.g. phosphorus) and source (e.g. final effluent from treatment works, agricultural pollution, etc.). Each risk layer uses openly available data to calculate a standardised risk value, which is added to a baseline catchment map. The Open Risk Map then determines how the level of risk varies throughout the river network in a catchment.

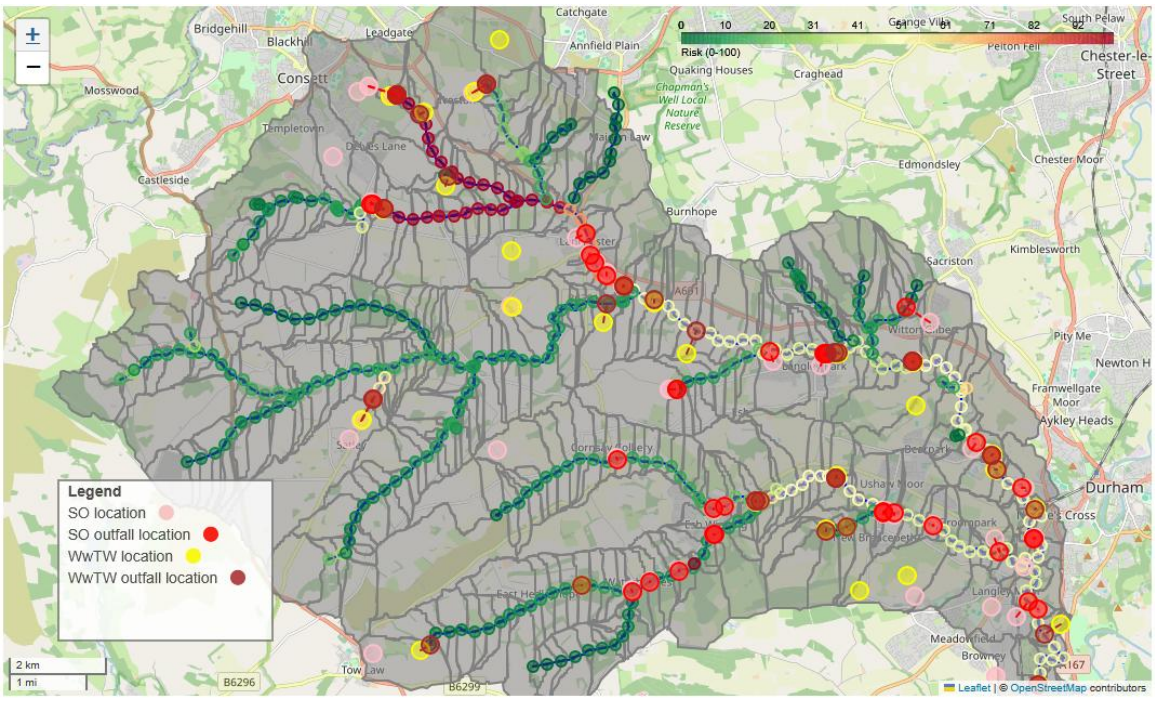

Figure 2-1 Risk distribution for the River Browney catchment, County Durham. The map shows orthophosphate-P risk [calculated as orthophosphate-P concentration as a percentage of the average of the UK TAG Revised orthophosphate-P standards for poor river health, 0.87 mg/l].

2.2 Open Poached Land Detection Model

The Open Poached Land Detection Model uses downscaled Sentinel-2 imagery to detect poached areas using remote sensing indices related to vegetation (NDVI) and bare soil (BSI).

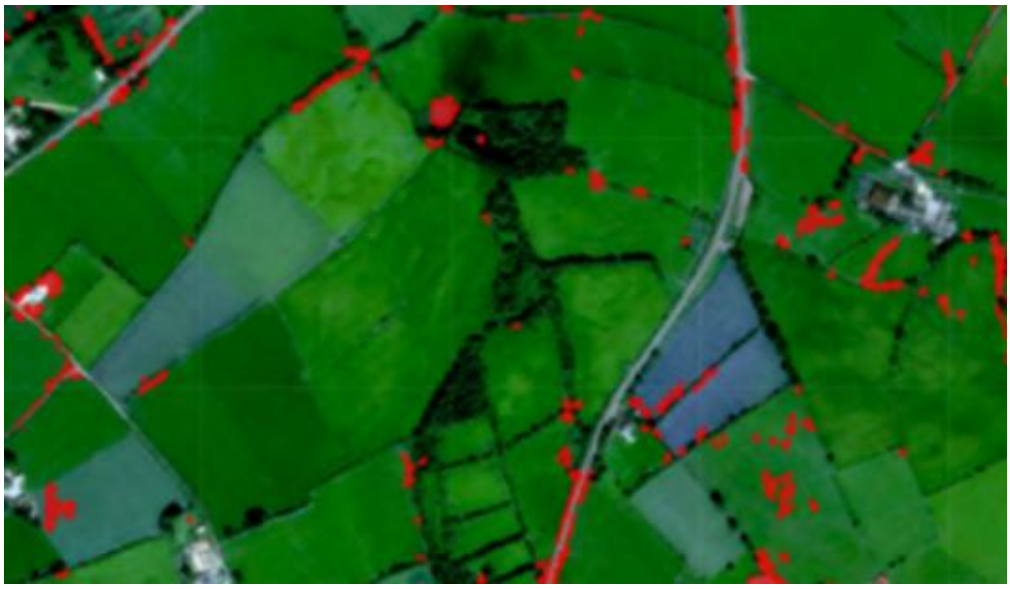

Figure 2-2 Poached land identified from satellite imagery

2.3 Open Slurry Tank detection Model

The Open Slurry Tank Detection model uses a YOLOv8 deep learning pipeline trained on public and custom data, to determine the locations of above-ground circular slurry tanks from detailed satellite imagery.

Figure 2-3 Slurry tank identified from satellite imagery

2.4 Open Bare Cropland Detection Model

The Open Bare Cropland Detection Model runs within Google Earth Engine (GEE), using Sentinel 2 imagery. It uses spectral indices such as Normalized Difference Vegetation Index (NDVI), Soil Adjusted Vegetation Index (SAVI), Bare Soil index (BSI), Normalized Difference Moisture Index (NDMI), and Normalized Difference Snow index (NDSI) to identify soil exposure within a user-specified time frame.

Figure 2-4 Bare cropland identified by the model from Sentinel-2 imagery within Google Earth Engine

2.5 Data inputs required

To run the models, various datasets are required (Table 2-1). All data used in the Open Risk Map model is open source and free to download. However, the Slurry Tank model requires paid for data, the Bare Cropland model runs in GEE, a paid for service (although the costs incurred for running the model are small) and the Poached Land model requires downscaled Sentinel-2 data (achieved using a paid for tool) and also runs in GEE.

Note that imagery used for the three detection models ideally needs to be cloud-free, which could potentially hinder near-real time deployment of the Poached Land and Bare Cropland models - although the repeat coverage time of Sentinel-2 data is typically less than 5 days, so there should be multiple potential datasets per month.

As the RDMAI models are open source, the user could amend the models to enable the use of alternative data, but this document focuses on the data sources used in the development of each model.

Table 2-1 Data requirements for the different models

RDMAI Model	Data Sources
Open Risk Map	The EA consents database needs to be manually downloaded into a stipulated folder. All other datasets are processed automatically by the provided script. • Rivers: Open River Network • Overflow spill data: EA Storm Overflows - Annual Returns • Consents data: EA Consents Database • Land use: CROME 2023 dataset • Hydrology of Soil Types (HOST) derived from this study. • Rainfall data: UK Met Office Rainfall Data • Roads and buildings: OS OpenMap - Local • Built-up areas: OS Open Built Up Areas • Farmscoper Data: ADAS Farmscoper Tool is used for agricultural and livestock runoff estimates via lookup tables.
Open Slurry Tank Detection	• Requires high-resolution (sub-1m) satellite or aerial imagery in GeoTIFF format
Open Poached Land Detection Model	• Sentinel-2 satellite imagery downscaled to 1m resolution (can be done using the paid for tool S2DR3) • Creation of a training dataset based on manual identification of poached land in downscaled imagery outside of the study area
Open Bare Cropland Detection Model	• Sentinel-2 satellite imagery (available within Google Earth Engine) • OSM Farmland and CROME crop datasets to constrain prediction areas

3 Workflow

3.1 Model set up and stakeholder requirements

Users will need to follow the instructions provided on the model GitHub repositories to download and run the models.

3.1.1 Open Risk Map Model

The RDMAI Open Risk Map model is designed to be highly accessible, with both the model and the required input data freely available. Its setup is straightforward and requires basic Python skills, such as cloning a GitHub repository, configuring a Conda environment, and working with Jupyter Notebooks. The model is largely "plug-and-play", enabling stakeholders to run analyses with minimal technical experience. Once the environment is prepared, running a catchment typically takes between 5 and 30 minutes on a standard laptop, depending on the catchment size. This makes the Risk Map model well-suited for users with limited coding experience who need quick and reliable outputs, that are repeatable for multiple catchments (Figure 2-1).

3.1.2 Open Poached Land Detection Model

Although the core of the Open Poached Land Detection Model runs within GEE, several preparatory steps must be completed outside of GEE. In particular, the high-resolution S2DR3 10b and 1 m imagery must first be generated using a Jupyter Notebook script. This script processes Sentinel2 data and exports the resulting files to Google Drive, from where they must be manually downloaded and uploaded into GEE as assets. This process can be time-consuming, especially as it requires selecting a suitable satellite acquisition date with minimal cloud cover. The script processes Sentinel-2 data and exports the resulting files to Google Drive, from where they must be manually downloaded and uploaded into GEE as assets.

The model also requires a training file (as this is not provided in the GitHub repository), which must be created by the user manually identifying poached areas from comparable imagery – a process that possibly only needs to be undertaken once within any organization, but would take several hours.

However, once the inputs are correctly configured and the model script edited for file paths etc., the model runs efficiently in GEE, typically requiring a few minutes for a WFD waterbody.

3.1.3 Open Slurry Tank detection Model

The model is delivered as a Jupyter Notebook intended to be largely 'plug and play' once the data path, YOLO model path, and a suitable virtual Python environment (with Python 3 installed) have been configured. Due to the potential need to downgrade to Python 3, one option is to switch the workflow to run in Google Colab, where both the data and the YOLO model are accessed directly from Google Drive, and the outputs written back to the same location.

The Slurry Tank model, executed in Google Colab, requires the data and YOLO model to be uploaded to Google Drive before processing. Once these are uploaded and the model set up, it takes up to an hour to run on 0.6m resolution data for a typical WFD waterbody.

The user also needs to purchase suitable resolution imagery for the area of interest. Due to the lifespan of slurry tanks, the exact date of the imagery is not important and so finding imagery with minimal cloud cover should not be a problem.

3.1.4 Open Bare Cropland Detection Model

The Open Bare Cropland Detection Model is the simplest of the three detection models to run, as once the datasets are loaded into GEE, the only necessary modifications to the code are specifying the file paths of the datasets and the time frame for the analysis (e.g. December 2024 – March 2025). The running of the model on average takes under 10 minutes for a typical WFD waterbody. However, the processing of the CROME and OSM input datasets can be time consuming depending upon the size of the area being modelled.

3.2 Identifying potential pollution hotspots

The RDMAI models are designed to complement existing approaches to identifying potential agricultural pollutant hotspots, rather than replace them. The models would thus be run alongside the items mentioned in Section 1.1, in particular, informing where within a catchment it would most likely be beneficial to undertake a catchment walk over.

The process would be to run one or more of the models and analyse the spatial output to see if there were obvious clusters of datapoints common between the models, indicating potential agricultural pressures. In particular, the Open Risk Map would identify pollutant 'hotspots', river reaches where agriculture was predicted to be the dominant source (for annual average pollution). The three detection models could then be used to identify if there was any significant extent of bare land or poaching in that area, or whether there were slurry tanks nearby, potentially indicating intensive livestock farming and associated pollutant pressures.

It would then be appropriate to undertake a catchment walk over in those areas to verify the model outputs to more fully understand the issues (as there are many other factors contributing to the presence and severity of agricultural pollution). One goal would be to make contact with landowners and begin dialogue to see if there were ways to help mitigate any pollutant issues identified, through for example, engagement with water company or government run schemes, including those identified in the RDMAI Waterbody Health Funding Handbook. The model outputs could potentially be used in such discussions to show the precited contribution of agriculture to river pollution or, using the outputs of the Poached Land model, identify specific areas where pollutant control measures should be considered.

One advantage of the Poached Land and Bare Cropland models is that they can be run for multiple dates and historically, helping to identify areas that are consistently shown as bare or poached, whereas the walkover would only identify issues present on the day it was undertaken.

3.3 Value Propositions

Whilst the RDMAI models are an additional data source, rather than a replacement for any existing data or task, they could facilitate improvements in the efficiency of catchment walk overs by allowing them to be more targeted at higher risk areas or areas with known issues.

The Open Risk Map should help target mitigation in areas where agriculture is predicted to be a significant contributor to the overall pollutant risk, or where mitigation of the overall risk would be most beneficial even though agriculture may not be the most significant contributor, thus improving the cost-efficiency of any catchment management scheme.

3.4 Adoption

Adoption of the Open Risk Map as an additional dataset to be considered as part of catchment management could be relatively straightforward given the model has no overheads, although the fact that it currently only includes phosphorus and ammoniacal-N might make it appear less relevant in catchments where these pollutants are not an issue. The main barrier to adoption is the ability to run the model and the associated proficiency in GIS and Python, which might necessitate an appropriately skilled person within the organization taking responsibility for running the model, rather than the person undertaking the walk over. However, this is no different to some of the existing tool-based approaches listed in Section 1.1 such as SciMap. One factor that would increase adoption is the addition of a user interface – as the RDMAI models are open source, anyone can attempt to create an interface around the underlying Python code. It may also be possible to include the models or model output within existing data packages and support networks, notably the Catchment Data User Group from the Catchment Based Approach.

Whilst there are comparable issues (and solutions) for adoption of the three Open Detection models, there are additional barriers to uptake, given the data needs of the models. For the Poached Land model, there is the upfront task of any organisation creating the required training dataset before they can use the model. For the Slurry Tank model, the cost of the detailed imagery data required (c. £10 km^{-2}) is significant given that a user can easily undertake the task manually using online mapping tools, and potentially learn more about the catchment in the process. The value of the Slurry Model would thus only be realised at a spatial scale where manual review would be prohibitive, or where the data was being purchased for reasons beyond the Slurry Model.

However, one set of factors that could increase the adoption of the Poached Land and Bare Cropland detection models is consistency and repeatability, allowing for an

assessment of historic issues, investigating change over time and the calculation of any associated metrics. These would be less reliable if based on catchment walk overs due to the subjective nature of human assessment and the difficulty and costs in repeat surveying the same areas year after year.

3.5 Limitations

The Open Risk Map is designed to provide an annual average estimation of risk across a catchment. Consequently, it does not intrinsically represent seasonal variations, rainfall-driven pollution spikes, or acute low-flow events, except insofar as these factors influence the annual average. Furthermore, the model relies on a rudimentary approximation of river flow to estimate in-stream pollutant dilution. Validation exercises have demonstrated that risk estimations could be considerably improved by substituting this approximation with measured flow data or an alternative dataset.

There are also limitations regarding the source data used to calculate pollutant loads in the Risk Map. Estimates for point source inputs, such as wastewater treatment works (WwTWs) and storm overflows (SOs), are derived from open data sources like the consents database. This reliance introduces a significant degree of uncertainty, as consent limits may not accurately reflect actual discharge volumes or frequencies. However, this uncertainty is not an intrinsic flaw of the model architecture and can be reduced by substituting open data with granular, proprietary water company data where available. Similarly, the estimation of agricultural diffuse pollution assumes a uniform distribution of livestock across a catchment, and does not account for variations in management or performance within a crop type, which oversimplifies the complex spatial heterogeneity of farming practices.

As partly implied by the name, the Open Slurry Tank detection model only identifies above-ground circular slurry tanks. However, these are only one way of storing manure, with significant proportions of slurry stored in lagoons or below ground tanks, various solid manure systems and some farms spreading manure directly. Thus the model only shows a proportion of all manure storage facilities and only highlights a proportion of the land where there is likely to be livestock.

The accuracy of the Poached Land model is 66% according to model documentation, but performed poorly in one of the two benchmarking catchments, with significant mis-identification of vegetated land as poached land and non-identification of actual poached land. These accuracy issues are in part related to the quality of the training dataset, which needs to be created by the user.

Further details on the models and their limitations can be found here.

4 Identifying hot spots in the Smallhope Burn

The Smallhope Burn (Figure 4-1), is a predominantly rural headwater catchment of the Browney catchment in the Wear Management Catchment in the Northumbrian Water operating area. The catchment has an area of around 40 km^2, with the towns of Consett located to the north, and Lanchester to the south. The waterbody is classified as being of Poor Ecological Status, with diffuse pollution from agricultural and rural land management cited as RNAGS. The waterbody is within both a Nitrate Vulnerable Zone and a Drinking Water Safeguard Zone. The burn is one of the initial focus areas of the 'Thriving Catchments' project set up by Northumbrian Water and the Rivers Trust, and engaging local partners.. Key objectives of the project included development of catchment solutions to remove phosphates from watercourses, including collaborating with local farmers and land managers to manage pollutants.

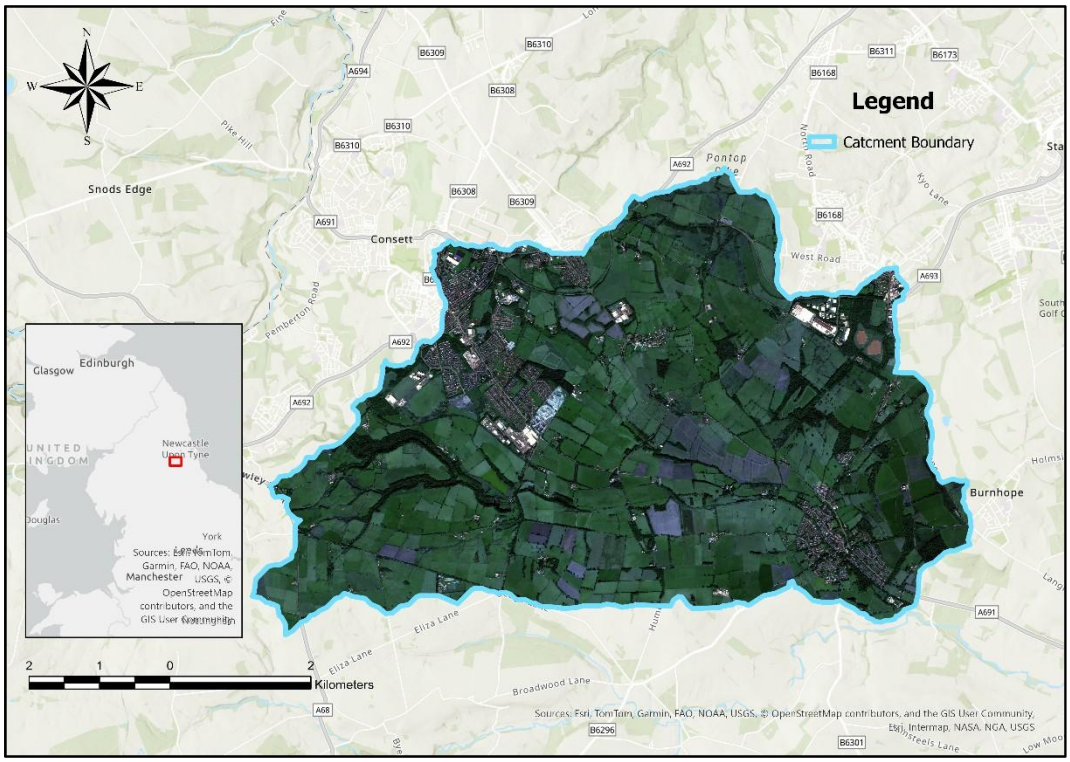

Figure 4-1 Location of WFD waterbody GB103024077330 (Smallhope Burn from Source to Browney, Stocke Water Body)

Figure 4-2 shows the output of the Open Risk Map for this catchment, with the tributaries to the west and north-west both being predicted to have a high phosphorus risk. The most significant contributor to these high risk areas is from WwTW (Figure 4-3), with the other contributions typically less important (SOs, urban, agricultural land use, and livestock ; Figure 4-4 to Figure 4-7). However, wherever WwTW are not the dominant contributing sector, livestock are predicted to be the main cause – notably in the western and northern tributaries (Figure 4-7).

The three Open Detection models were then used to try and identify any potential issues in those areas where livestock (or agriculture) were found to be important for phosphorus risk:

- The Bare Cropland detection model identifies a number of bare fields in the winter of 2023/24 (Figure 4-9), which could be a potential source of pollution and contribute to the elevated phosphorus risk – repeated application of the model for other winters would show whether there was any consistency in the extent or location of bare cropland. If the land could be managed such that it was not left bare over-winter (e.g. through the use of cover crops), this might help reduce agricultural pollution – although the risk map suggests that agricultural land use is not a major contributor to overall phosphorus risk.
- The Poached Land detection model found small areas of poaching all across the catchment (Figure 4-8), suggesting that there are no particular localised areas or clusters of hot spots that could potentially exacerbate any agricultural pollution occurring and would be worthy of investigation. This is not necessarily a negative result, as the model is suggesting that poaching may be a pervasive issue across the catchment. As per the bare cropland model, repeated application of the model for other time periods could help to identify persistent areas of poaching.
- Application of the Open Slurry Tank detection model found no slurry tanks in the catchment, and thus in this instance is not able to help focus any catchment investigation (ground-truthing in the catchment also found no slurry tanks, although there is a slurry lagoon north-east of Lanchester and plenty of solid manure heaps).

The Open Risk Map was able to identify a few tributaries where agriculture, particularly livestock, were a potentially large contributor to the phosphorus risk. However, partly due to the agriculture practices within in the catchment, the Open detection models were not able to provide insight into where around those tributaries any particular hot spots might be and/or it would be beneficial to undertake further investigation in-person.

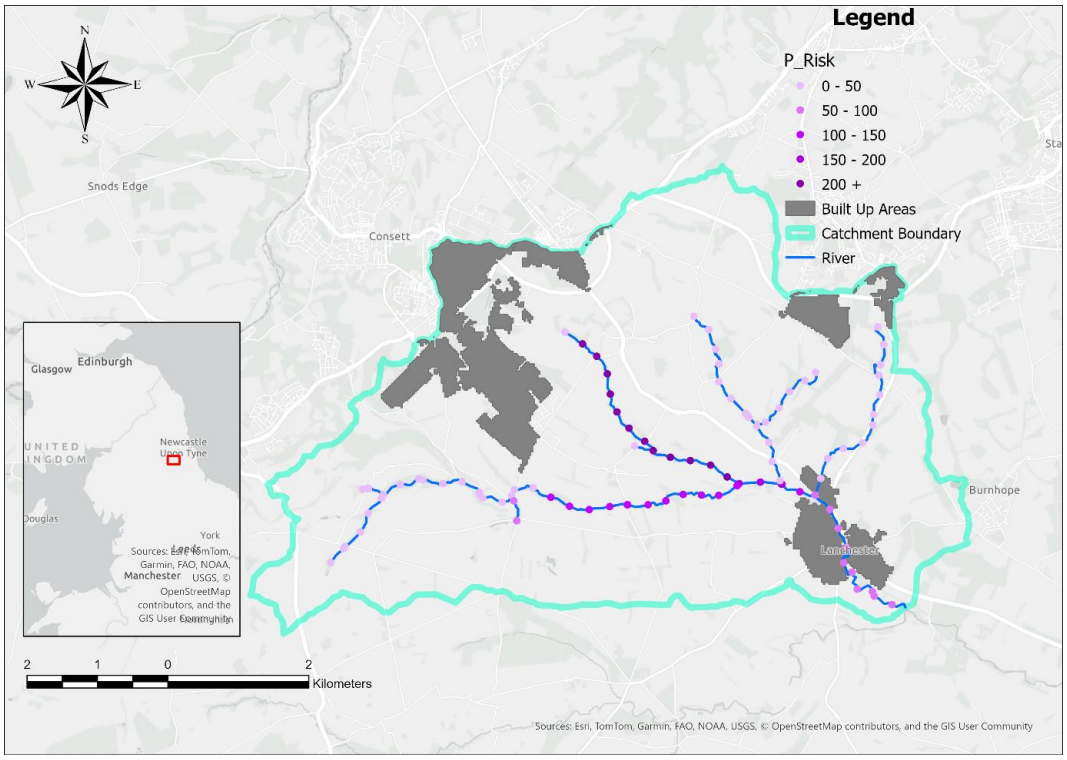

Figure 4-2 the RDMAI Open Risk Map estimated phosphorus risk for the Smallhope Burn waterbody (GB103024077330). Risk is calculated as orthophosphate-P concentration as a percentage of the average of the UK TAG Revised orthophosphate-P standards for poor river health, 0.87 mg/l.

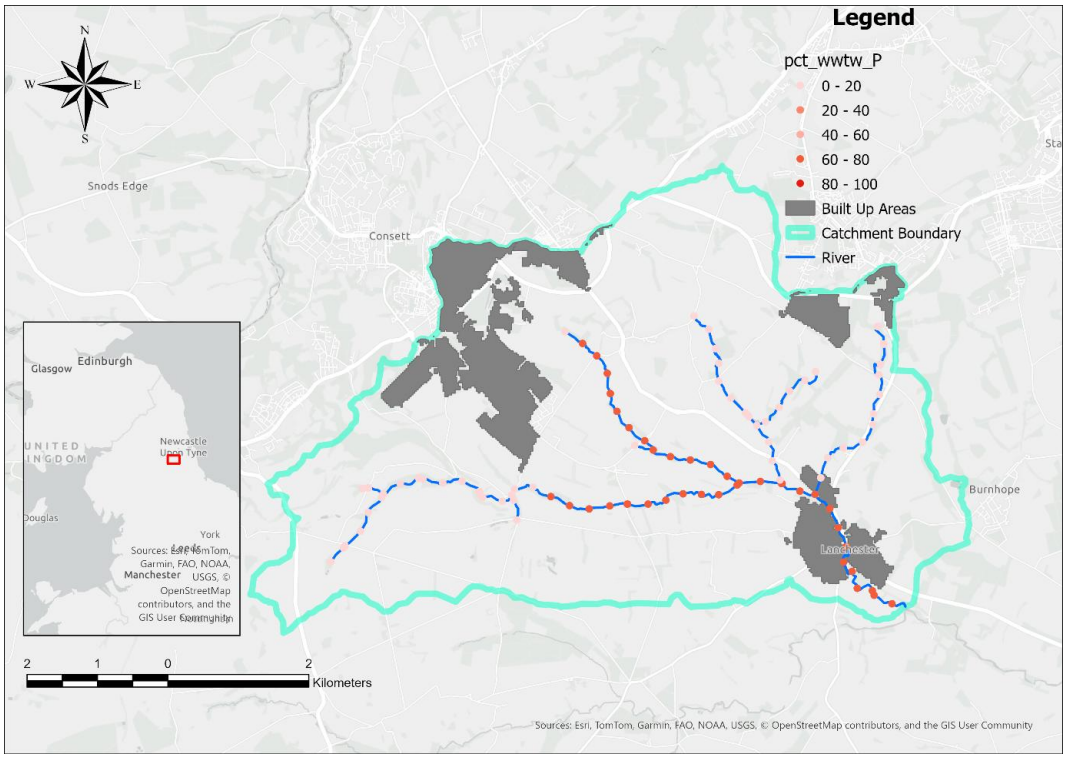

Figure 4-3 Wastewater treatment works contribution (as a percentage) to phosphorus risk as calculated by the Open Risk Map model for the Smallhope Burn waterbody (GB103024077330)

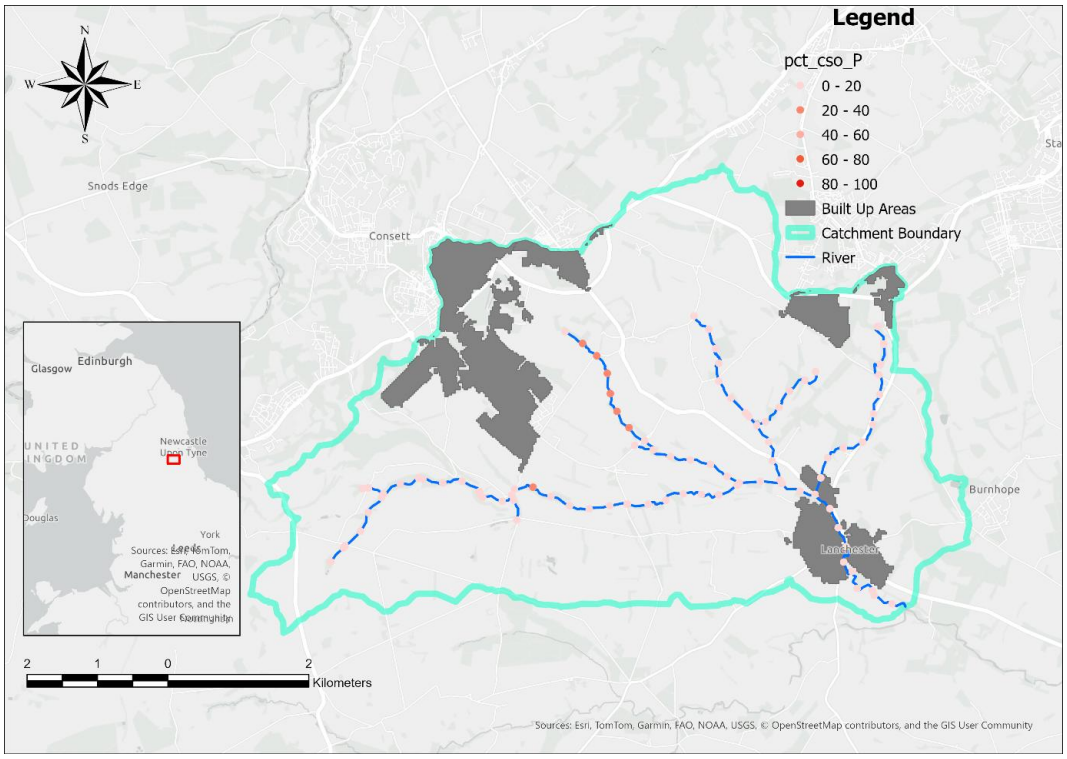

Figure 4-4 Storm overflow contribution (as a percentage) to phosphorus risk as calculated by the Open Risk Map model for the Smallhope Burn waterbody (GB103024077330)

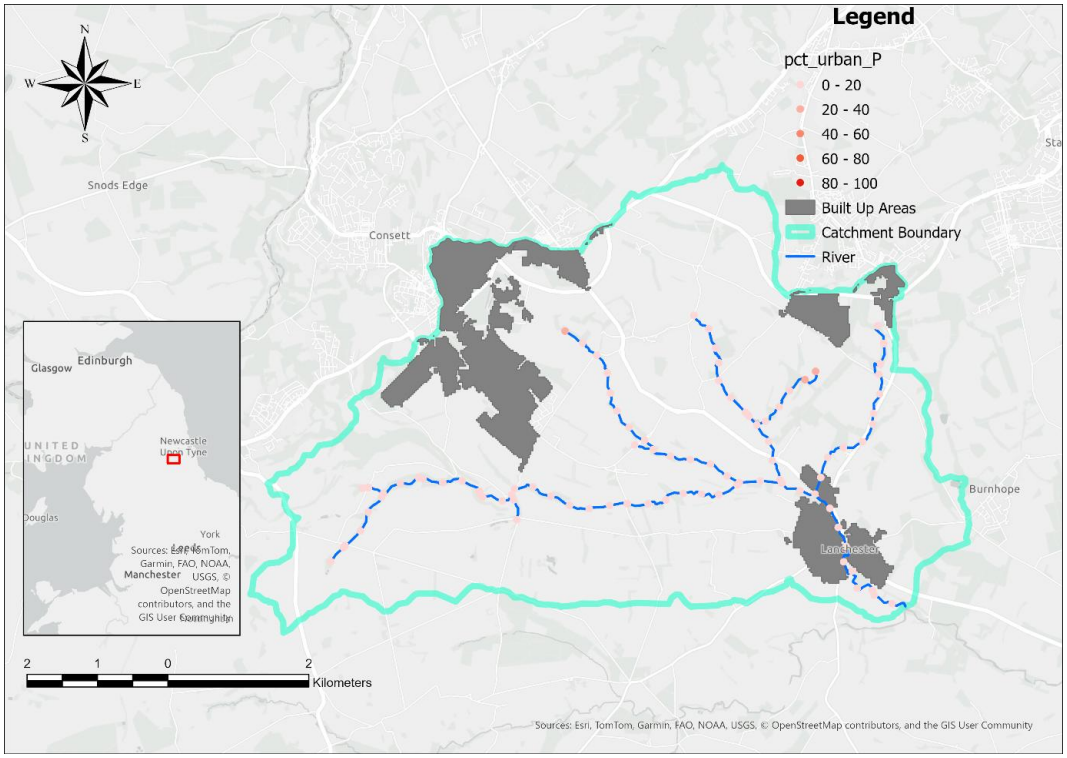

Figure 4-5 Urban contribution (as a percentage) to phosphorus risk as calculated by the Open Risk Map model for the Smallhope Burn waterbody (GB103024077330)

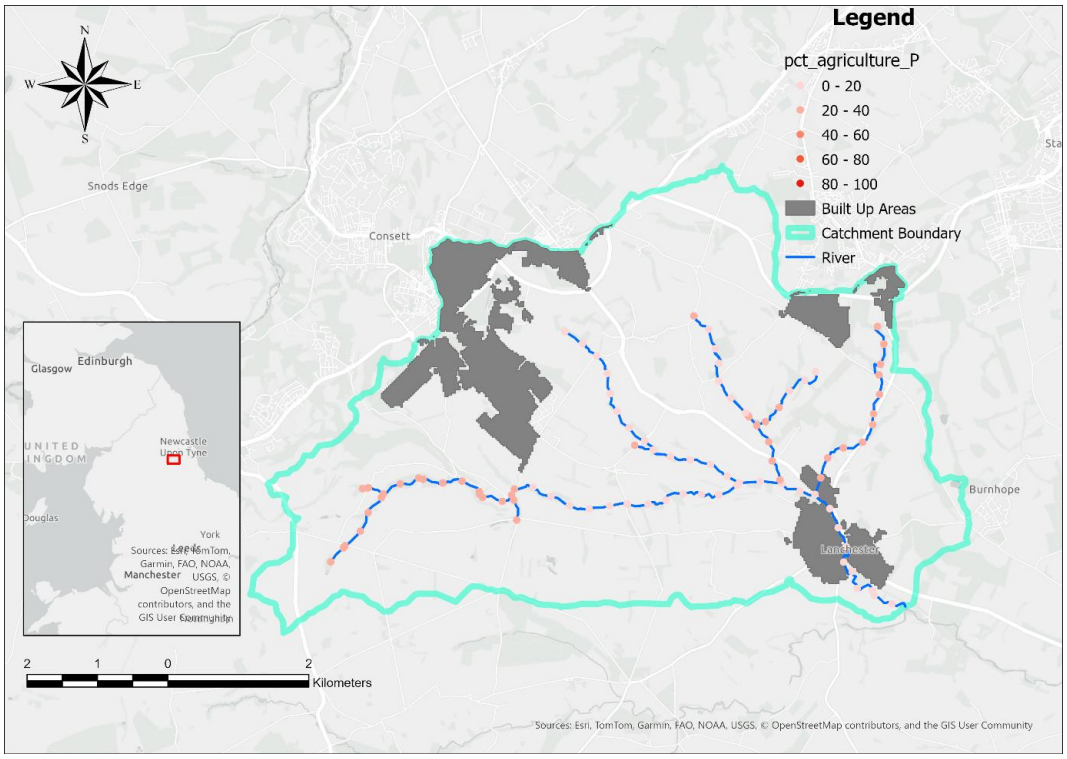

Figure 4-6 Agricultural land contribution (as a percentage) to phosphorus risk as calculated by the Open Risk Map model for the Smallhope Burn waterbody (GB103024077330)

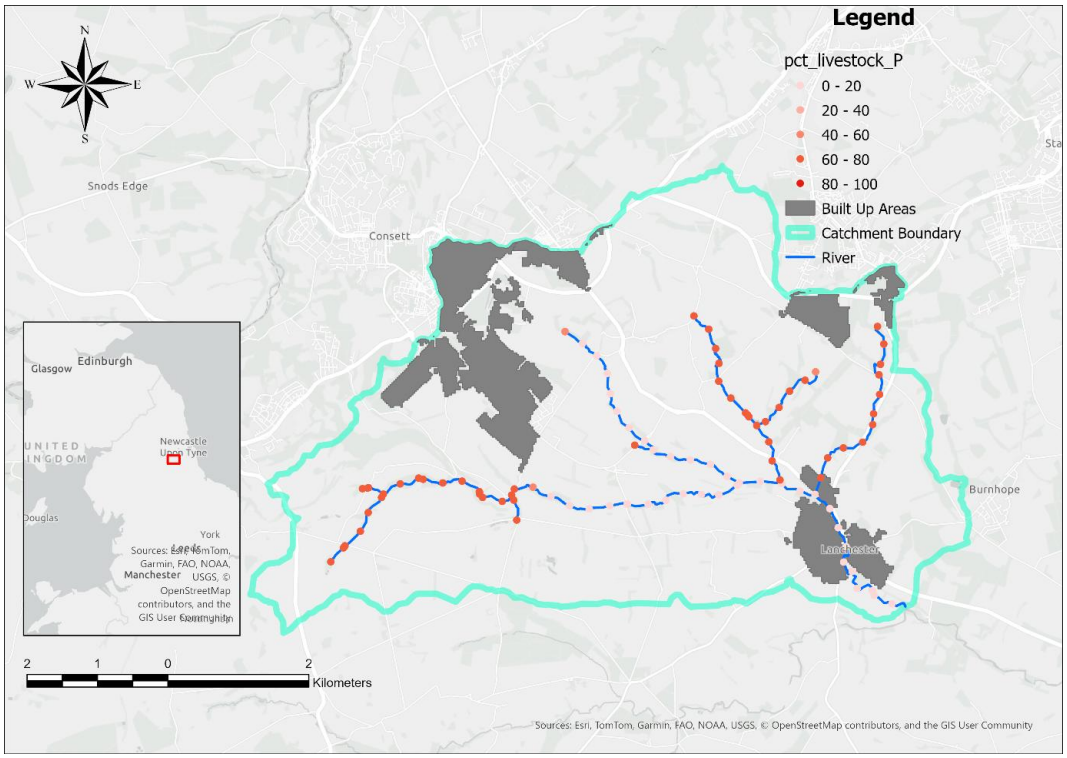

Figure 4-7 Livestock contribution (as a percentage) to phosphorus risk as calculated by the Open Risk Map model for the Smallhope Burn waterbody (GB103024077330)

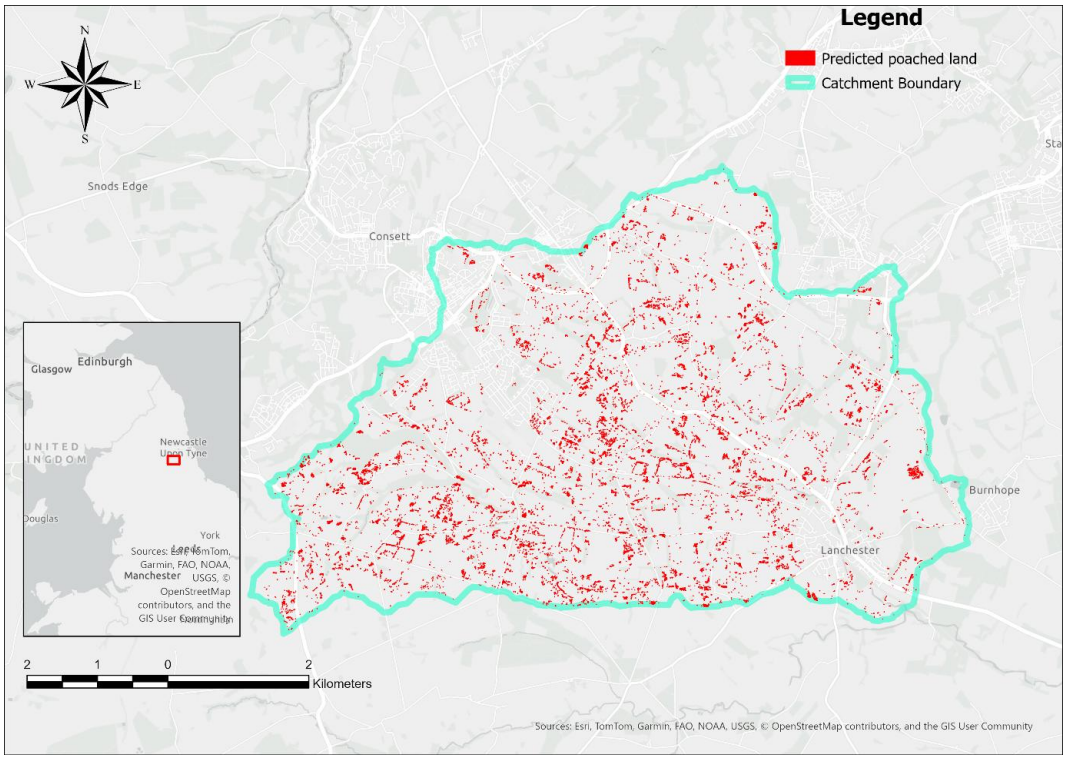

Figure 4-8 Areas of poaching identified by the Open Poached Land detection model for the Smallhope Burn waterbody (GB103024077330)

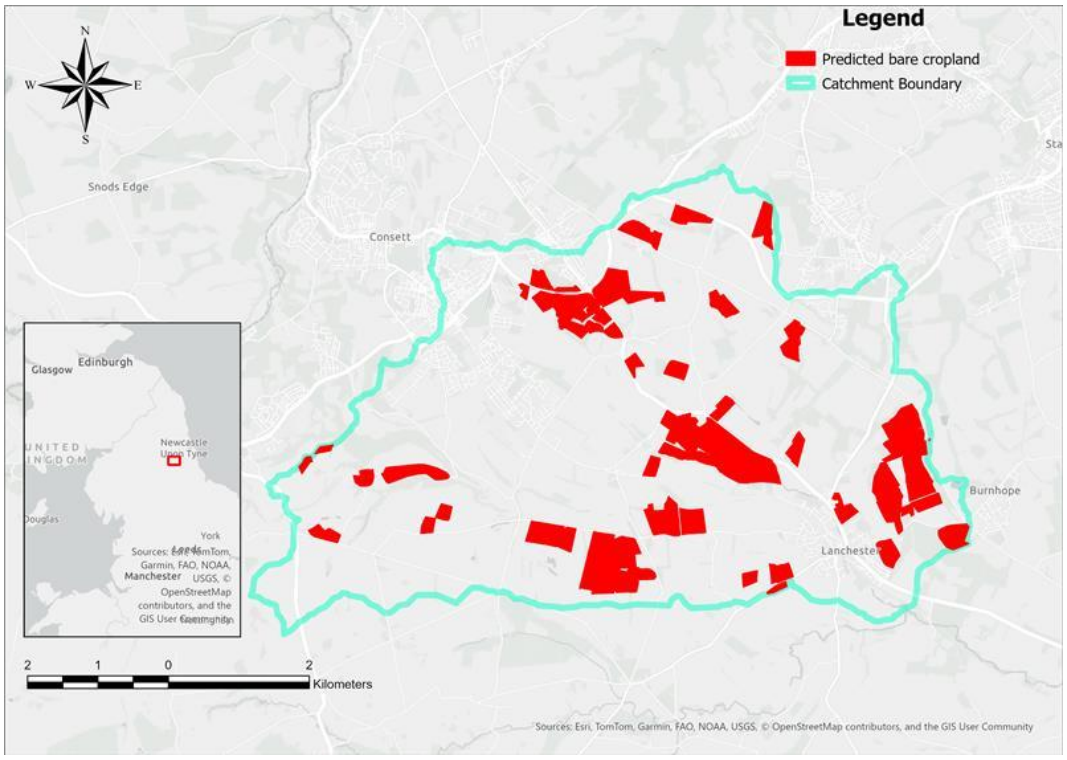

Figure 4-9 Areas identified by the Open Bare Cropland detection model for the Smallhope Burn waterbody (GB103024077330)

5 Next steps and opportunities

Given that the Open Slurry Tank model does not identify other forms of manure storage, one key area opportunity for further development of the model is to expand its capabilities so that it can. Also, the model relies upon relatively expensive detailed imagery datasets, so part of the development could review the potential to use alternative cheaper datasets, such as the S2DR3 downscaled Sentinel 2 data used in the Poached Land model.

The Poached Land model does not currently include the required training file, so a repository for user-created training files may be beneficial and could potentially lead to improved model performance as both the amount of training data, and variation in the landscapes for which training data has been derived, increases. The Poached Land model also only predicts land as either poached or not poached, when there is the potential to use the indices calculated as part of the model to calculate a 'poaching index'. This would allow the users to assess whether there is a better agreement with observed areas of poaching above some threshold, or even just to target the higher index areas as part of a catchment walkover.

Turning the existing Python scripts into a more user-friendly piece of software or simply adding a user-interface would likely result in increased adoption of the tools. Given the relative similarity of the three detection models (particularly if they could all be run from the same imagery data), there could be merit in combining them within the same interface.

References

Defra (2018), A Green Future: Our 25 Year Environment Plan. 151pp.

Environment Agency (2021) Agriculture and rural land management: challenges for the water environment. 21pp.

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